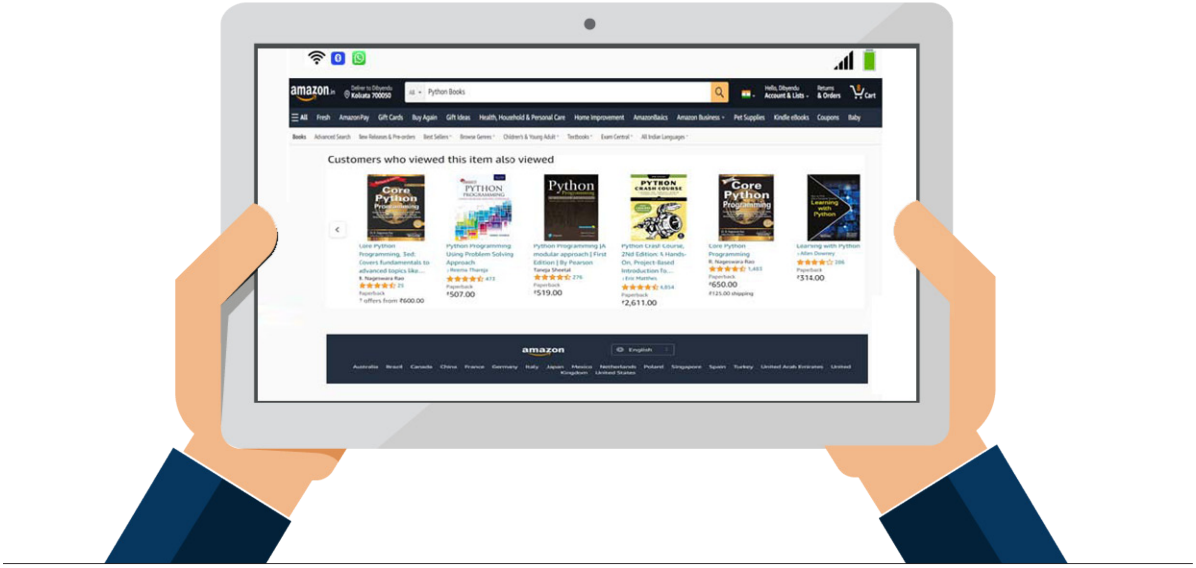
Recommender Systems: An Overview of the Mathematics

This two-part series of articles explains and demonstrates how to implement a recommender system for an online retail store using Python. In this first part of the series, the focus is on the theory behind such a system. The implementation will be covered in the second part.



Recommender systems are now widely used to personalise your online experience, advising you on what to buy, where to dine, and even who you should be friends with. People's preferences vary, yet they tend to follow a pattern. They have a tendency to like things that are comparable to the other things they enjoy, and they have tastes that are similar to the tastes of the people they know. Recommender systems attempt to capture these trends to forecast what users might enjoy in the future. E-commerce sites, social networks, and online news platforms actively implement their own recommender systems to assist customers in making product choices more efficiently.

You would have observed that Netflix and Amazon Prime recommend content that is similar to what you watched previously. So, how does this happen? It's because of a machine learning based recommendation engine that figures out how similar the videos are to other things you like, and then it predicts your choices.

In 2006, Netflix announced a competition to develop an algorithm that would "significantly increase the accuracy of forecasts about how much someone will enjoy a movie based on their movie tastes." Since then, Netflix has expanded its use of data to include personalised ranking, page generation, search, picture selection, messaging, and marketing, among other things. The Netflix Recommendation Engine (NRE), its most effective algorithm, is made up of algorithms that select content based on each individual user profile. It's so accurate that customised recommendations drive 80 per cent of Netflix's viewer engagement.

Netflix isn't the only business that uses a recommender system. Amazon, LinkedIn, Spotify, Instagram,